Multimedia Signal processing

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1. **Question (8 pts)**:
   Consider a continuous source X, whose p.d.f. is given by

$$p_X(x) = e^{-x}, \quad x \geq 0 \tag{1}$$

   (a) Design a quantizer based on compander-expander. Use 1 bits to represent each symbol. Show the equivalent non-uniform quantizer in the original signal domain.

   (b) Compute the entropy of the output symbols of the quantizer.

   Solution

   (a) In order to design the nonlinear transformation used in the compander-expander quantizer, we need to write the following expression, which denotes the c.d.f. of the source X

$$y = T(x) = \int_{-\infty}^{x} p_X(a)da = \int_{0}^{x} e^{-a}da = \left[-e^{-a}\right]_0^x = 1 - e^{-x} \tag{2}$$

   The output y is quantized with a uniform quantizer whose intervals and reconstruction levels are

$$[0, 1/2) \rightarrow y_1 = 1/4 \tag{3}$$
$$[1/2, 1) \rightarrow y_2 = 3/4 \tag{4}$$

   The equivalent non-uniform quantizer in the original signal domain is obtained by inverting $T(x)$. That is,

$$x = T^{-1}(y) = -\log(1-y) = \log\frac{1}{1-y} \tag{5}$$

   The output y is quantized with a uniform quantizer whose intervals and reconstruction levels are

$$[0, \log(2)) \rightarrow y_1 = \log(4/3) \tag{6}$$
$$[\log(2), \infty) \rightarrow y_2 = \log(4) \tag{7}$$

   (b) The probability of observing y_k is given by

$$\int_{a_k}^{a_{k+1}} p_X(a)da \tag{8}$$

   Hence,

$$p_1 = p(y_1) = \int_{0}^{\log 2} e^{-a}da = \frac{1}{2} \tag{9}$$
$$p_2 = p(y_2) = \int_{\log(2)}^{\infty} e^{-a}da = \frac{1}{2} \tag{10}$$

The entropy is

$$H(y_k) = -p_1 \log_2(p_1) - p_2 \log_2(p_2) = 1 \text{ bits / symbol} \tag{11}$$

Hence, there is no coding gain from entropy coding the output symbols of the quantizer.

2. **Question (5 pts)** Discuss the utility, in terms of coding efficiency, of colorspace transformations in lossless and lossy image coding.

   Solution

   See slide set `MMSP_2.2_Image coding`.

3. **Question (5 pts)**

   (a) Illustrate the block diagram of a hybrid DCT/DPCM video codec.

   (b) Detail the pseudocode describing the sequence of processing steps applied at the encoder.

   Solution

   See slide set `MMSP_2.3_Video_Coding`.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**

   (a) Let X and Y denote two continuous sources with memory, and $Q_X(\cdot)$, $Q_Y(\cdot)$ the Lloyd max quantizers designed based on the corresponding p.d.f.'s. Let $\hat{X}$ and $\hat{Y}$ the sources describing the output of the quantizers.

   - [T] [F] $\hat{X}$ and $\hat{Y}$ are discrete memoryless sources.
     FALSE. The sources are discrete (since their symbols are the output levels of the quantizers). However, the memory of the source is preserved after quantization.
   - [T] [F] If X and Y have the same variance, but different p.d.f.'s (e.g., one is Gaussian and the other Laplacian), $Q_X(\cdot) = Q_X(\cdot)$
     FALSE. The design of the Lloyd-Max quantizer depends also on the shape of the pdf.
   - [T] [F] $H(\hat{X}) \leq H(X)$.
     FALSE. The entropy is undefined for continuous sources.

   (b) Consider an audio coding system

   - [T] [F] The analysis filterbank uses the same filters as the perceptual model.
     FALSE. The perceptual model can use a different set of filters, since it is not defined by the standard.
   - [T] [F] The processing steps performed at the encoder are specified by the selected standard.
     FALSE. The standard specifies only the decoder.
   - [T] [F] MDCT does not achieve perfect reconstruction.
     FALSE. MDCT does achieve perfect reconstruction.
   - [T] [F] The impact of frequency masking decreases with the distance from the frequency of the masker.
     TRUE. Frequency masking is more pronounced at the frequency corresponding to the masker, and decreases with the distance.

   (c) Consider a transform coding system

   - [T] [F] The coding efficiency always achieves that of vector quantization.
     FALSE. Transform coding is able to capture only linear dependencies between data. Conversely, vector quantization is able to capture also non-linear statistical dependencies.
   - [T] [F] Under optimal bit allocation, more bits are assigned to coefficients having larger variance.
     TRUE.
   - [T] [F] The DCT always maximizes the transform coding gain.
     FALSE. The KLT does. However, the DCT is often a good approximation of the KLT.

5. **Question (Matlab - 10pts)**

   Load the files *f1.mat* and *f2.mat*, which contain, respectively, the first and the second frame of the *Foreman* sequence.

   (a) Compute the entropy of the two frames.

   (b) Which coding scheme between the following is the best choice for lossless coding the second frame?

   - Intra-frame coding, in which each pixel is predicted using the pixel located on top of it
   - Inter-frame coding, in which each pixel is predicted using the same pixel in frame 1

   Solution

```
clear all;
close all;
clc;

load f1;
load f2;


%compute entropy of first frame
h1 = hist(f1(:),0:255);
p1 = h1./sum(h1);
p1 = p1(p1~=0);
H1 = -sum(p1.*log2(p1))

%compute entropy of second frame
h2 = hist(f2(:),0:255);
p2 = h2./sum(h2);
p2 = p2(p2~=0);
H2 = -sum(p2.*log2(p2))


%intra-encoding: compute joint entropy between adjacent pixels in the
%second frame
f2_ref = f2(1:end-1,:);
f2_code = f2(2:end,:);

h22 = hist3([f2_ref(:) f2_code(:)],[256 256]);
p22 = h22./sum(sum(h22));
p22 = p22(:);
p22 = p22(p22~=0);
H22 = -sum(p22.*log2(p22))

%inter-encoding: compute joint entropy between the two frames
h12 = hist3([f1(:) f2(:)],[256 256]);
p12 = h12./sum(sum(h12));
p12 = p12(:);
p12 = p12(p12~=0);
H12 = -sum(p12.*log2(p12))
```